EXP-8

CODE:

```
clc;
clear;
close all;

Fs = 1000;
N = 20;
L = 128;

% FIR filter coefficients
h = fir1(N,0.3,hamming(N+1));

% Unit impulse input
x = [1 zeros(1,L-1)];

% Direct realization
y_direct = filter(h,1,x);

% Cascade realization using second-order sections
[sos,g] = tf2sos(h,1);
y_cascade = g*x;

for k = 1:size(sos,1)
    y_cascade = filter(sos(k,1:3),sos(k,4:6),y_cascade);
end
```

```
% Check coefficient symmetry
symmetry_error = max(abs(h-fliplr(h)));

% Frequency and phase response
[H,f] = freqz(h,1,1024,Fs);

% Plot impulse responses
figure;
stem(0:L-1,y_direct,'b');
hold on;
stem(0:L-1,y_cascade,'r--');
grid on;
xlabel('Samples');
ylabel('Amplitude');
title('Direct vs Cascade FIR Realization');
legend('Direct','Cascade');

% Phase response
figure;
plot(f,unwrap(angle(H)));
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (Radians)');
title('Linear Phase Response');

% Display results
disp('FIR Coefficients:');
```

```
disp(h);

disp('Cascade SOS Coefficients:');
disp(sos);

fprintf('Symmetry Error = %g\n',symmetry_error);
fprintf('Maximum Output Difference = %g\n', ...
       max(abs(y_direct-y_cascade)));
fprintf('Linear Phase Group Delay = %.1f samples\n',N/2);
fprintf('Group Delay = %.4f seconds\n',(N/2)/Fs);
```

OUTPUT:

```
FIR Coefficients:
  Columns 1 through 12

         0    0.0029    0.0063    0.0038   -0.0124   -0.0343
-0.0319    0.0265    0.1379    0.2513    0.2996    0.2513

  Columns 13 through 21

    0.1379    0.0265   -0.0319   -0.0343   -0.0124    0.0038
0.0063    0.0029         0

Cascade SOS Coefficients:
         0    1.0000         0    1.0000         0         0
    1.0000   -3.4877    3.3363    1.0000         0         0
    1.0000    0.4529    1.2411    1.0000         0         0
```

1.0000	1.4121	1.0000	1.0000	0	0
1.0000	0.8801	1.0000	1.0000	0	0
1.0000	1.9754	1.0000	1.0000	0	0
1.0000	1.7823	1.0000	1.0000	0	0
1.0000	-0.1692	1.0000	1.0000	0	0
1.0000	0.3649	0.8057	1.0000	0	0
1.0000	-1.0454	0.2997	1.0000	0	0

Symmetry Error = 0

Maximum Output Difference = 3.10862e-15

Linear Phase Group Delay = 10.0 samples

Group Delay = 0.0100 seconds



